

Live Lab Validation Guide & Runbook

This runbook guides you through triggering live attacks from within your virtual lab network and verifying end-to-end NDR ingestion, dual-path detection, and automated containment.

1. Network Topology Checklist

- [] **OPNsense Firewall VM:** Running with `em0` (WAN), `em1` (LAN 192.168.10.1/24), `em2` (Quarantine 192.168.99.1/24).
 - [] **Monitor VM (Ubuntu 22.04 LTS):** Running with `eth0` (Mgmt: 192.168.10.20) and `eth1` (SPAN / Promiscuous Tap).
 - [] **Attacker / Target Workstations:** Running on VLAN 10 (192.168.10.0/24).
-

2. Triggering Lab Attacks

A. Network Reconnaissance & Port Scanning (MITRE ATT&CK T1046)

From your Attacker VM:

```
# Run aggressive Nmap SYN scan against internal target
sudo nmap -sS -p 1-1000 -T4 192.168.10.50
```

- **Expected Suricata Alert:** `SID 3000001 (NDR T1046 - Rapid TCP SYN Portscan Detected)`
 - **Expected Classifier:** `PORT_SCAN (Confidence > 90%)`
 - **Expected Action:** Source IP added to `ndr_blocked_ips` table on OPNsense.
-

B. High-Volume TCP SYN Flood (MITRE ATT&CK T1498.001)

From your Attacker VM:

```
# Generate high-rate SYN flood using hping3
sudo hping3 -S -p 80 --flood --rand-source 192.168.10.50
```

- **Expected Suricata Alert:** `SID 3000002 (NDR T1498.001 - High-Rate TCP SYN Flood Target Overload)`
 - **Expected Classifier:** `DDOS_FLOOD`
 - **Expected Action:** `BLOCK_AND_ISOLATE`
-

C. DNS Tunneling / Data Exfiltration (MITRE ATT&CK T1071.004)

From your Attacker VM:

```
# Simulate DNS exfiltration queries using iodine or script
python scripts/generate_test_traffic/generate_attacks.py
```

- **Expected Entropy:** `dns_subdomain_entropy > 3.8`
 - **Expected Decision:** `BLOCK_AND_ISOLATE`
-

D. Atomic Red Team C2 Simulation (MITRE ATT&CK T1071.001)

From your Workstation:

```
# Execute Atomic Red Team HTTP C2 test
python scripts/generate_test_traffic/atomic_runner.py --technique T1071.001 --target 203.0.113.88
```

- **Expected Classifier / Autoencoder:** `C2_BEACONING / Loss > Threshold`
 - **Expected Action:** `QUARANTINE_VLAN` (VLAN 99)
-

3. Verifying Results in Live Lab

1. Check OPNsense live firewall aliases:

```
https://192.168.1.1/ui/firewall/alias -> inspect ndr_blocked_ips .
```

2. Inspect SIEM alerts dispatched to Wazuh Manager:

```
http://localhost:5601 or /var/ossec/logs/alerts/alerts.json .
```